Who Gets Shade? Mapping Urban Tree Canopy & Heat Inequality

in Southern Denton County, Texas, Using Airborne LiDAR

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USGS 3DEP LiDAR · NAIP · Landsat 8/9 · ACS

1 Why it matters

North Texas summers are getting hotter, and heat is the deadliest weather hazard in the U.S. Trees are a city's natural air-conditioning: their **shade and transpiration** cool streets, roofs and people. But tree canopy is not shared equally.

Airborne LiDAR measures the 3-D height of every tree and building, so we can map real canopy, not just "green pixels", across a whole county at 2 m resolution.

Why LiDAR, not just satellite "greenness"?

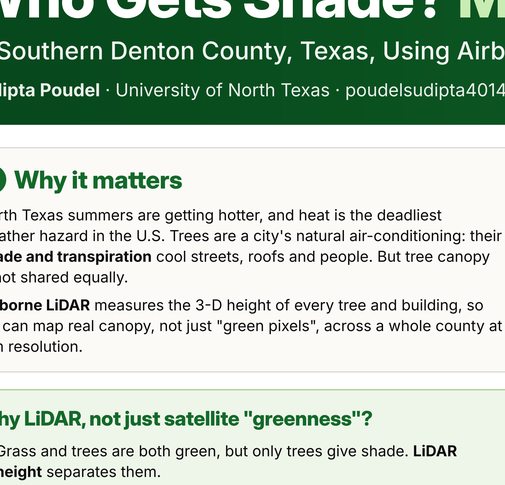
- Grass and trees are both green, but only trees give shade. **LiDAR height** separates them.
- Measures every tree **crown at 2 m**, so it catches single street trees.
- Gives tree **height** (mean 9.6 m here) and 3-D structure.

2 Research questions

1. How much **tree canopy** does each neighborhood have, measured with LiDAR?
2. Do low-canopy neighborhoods have **hotter** summer surfaces?
3. Are **"shade deserts"** (low canopy + high heat) concentrated in lower-income neighborhoods?

3 Study area & data

Study Area — Southern Denton County, Texas



287 census block groups · 535,601 residents · 742 km². Limited to areas fully covered by the 2018 USGS 3DEP LiDAR (north Denton incl. UNT main campus not yet covered).

Dataset	Res.	Use
USGS 3DEP airborne LiDAR (2018)	2 m	Height above ground
USDA NAIP imagery, RGB-NIR (2018)	0.6 m	NDVI: trees vs. roofs
Landsat 8/9 surface temp., 17 clear summer scenes (2018–22)	30 m	Heat
Census ACS 5-yr (2018–22), TIGER/Line	Block group	Income, population

4 Methods

LiDAR point cloud
Height above ground (HAG), 2 m grid

NAIP NDVI
Greenness from near-infrared

Landsat LST
Cloud-masked, per-pixel summer median

↓

Classify every 2 m cell

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tree     = HAG ≥ 3 m AND NDVI ≥ 0.05
building = HAG ≥ 3 m AND NDVI < 0.05
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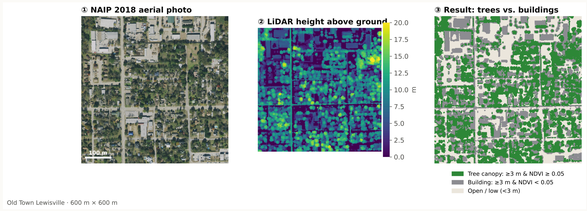
NDVI threshold = valley of the bimodal NDVI histogram of tall cells (roofs ≈ −0.05, crowns ≈ +0.18)

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Zonal statistics
% canopy, % buildings, mean LST per block group

Statistics
Correlation, OLS regression, income quintiles

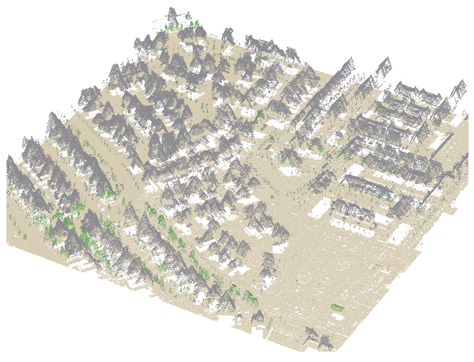
Shade deserts
Lowest-25% canopy AND hottest-25% LST



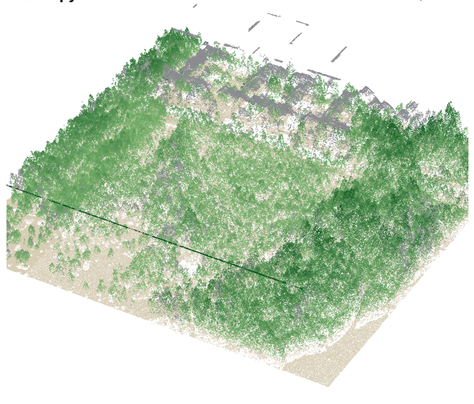
LiDAR + NAIP classification in Old Town Lewisville. Spot-check vs. raw LiDAR points on a 1 km tile: 82% agreement on ≥3 m cells.

5 Seeing the difference in 3-D

Shade desert
Canopy 4% · Summer LST 49.0 °C · Median income $125k



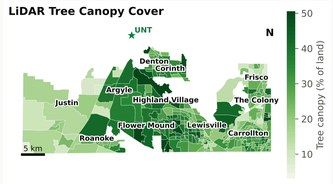
Leafy & cool
Canopy 59% · Summer LST 42.3 °C · Median income $66k

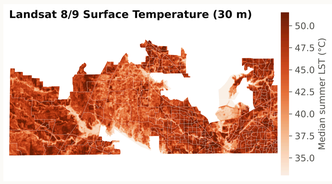


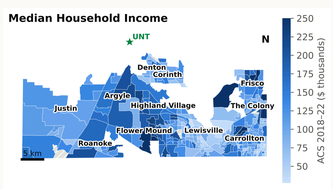
Tree canopy (≥3 m & green) · Buildings (≥3 m & not green) · Ground / low objects

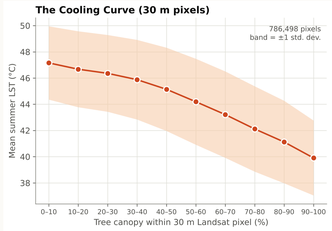
Raw 2018 LiDAR points (≈1.5–1.8 million per 600 × 600 m view). Same county, same summers: a **6.7 °C** difference in surface temperature.

6 Mapping canopy, heat & income



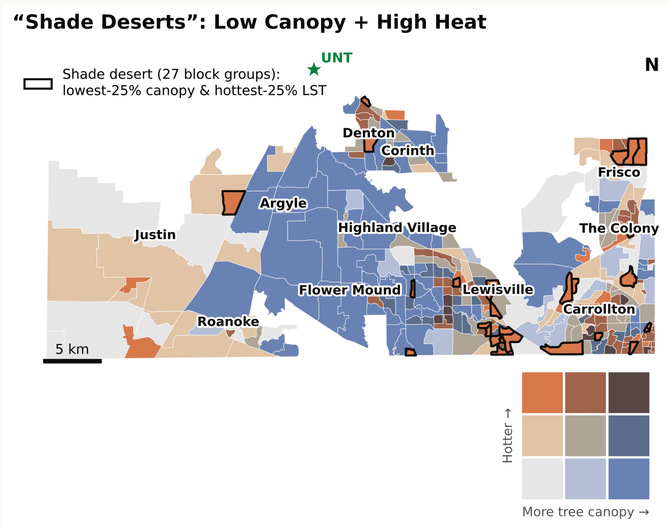






Overall LiDAR canopy cover: **26.5%** (block groups range 3–59%). Mean summer surface temperature ranges 39–51 °C.

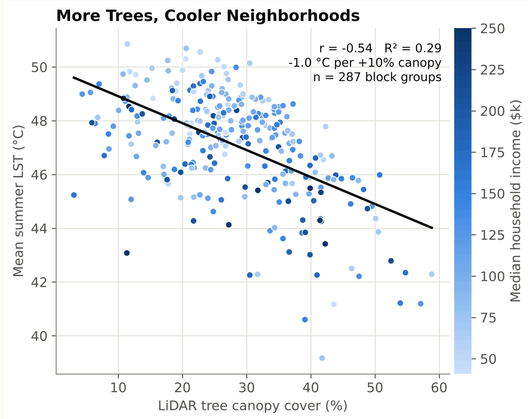
7 Where are the shade deserts?



27 block groups are shade deserts, home to **48,677 people**: average canopy 14.7% vs. 26.5% overall, and 49.4 °C summer surfaces. Many sit along the dense I-35E corridor and in fast-growing new subdivisions in the east.

City (≥5 block groups)	Block groups	Avg. canopy	Avg. LST	Shade deserts	Residents in them
Lewisville	66	27%	47.8 °C	11	14,389
Carrollton	47	26%	47.7 °C	4	9,023
Flower Mound	38	37%	45.9 °C	2	2,980
Dallas	21	28%	48.8 °C	3	3,966
The Colony	20	27%	47.0 °C	1	1,609
Frisco	17	18%	47.2 °C	3	9,733
Denton	14	26%	47.8 °C	2	5,082
Unincorporated	13	27%	45.6 °C	1	1,895
Corinth	12	33%	46.8 °C	0	0
Fort Worth	8	15%	46.9 °C	0	0
Roanoke	5	24%	47.1 °C	0	0

8 Results

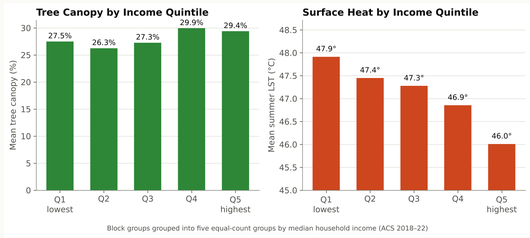


-1.0 °C
per +10% tree canopy (r = -0.54, p < 0.001)

5.2 °C
hotter: 30 m pixels with <10% vs. ≥50% canopy

1.9 °C
hotter: lowest- vs. highest-income fifth of neighborhoods

−26%
median income in shade deserts ($85k vs. $116k)



Canopy explains heat even after accounting for buildings and population density (-1.0 °C per +10% canopy; model R^2 = 0.72). Canopy itself differs only weakly by income (ρ = 0.13); dense, built-up development drives much of the income–heat gap.

9 Takeaways

- **Trees cool:** every 10% more canopy ≈ 1 °C cooler surfaces.
- **Heat is unequal:** lower-income neighborhoods are ~2 °C hotter; shade deserts have 26% lower incomes.
- **New subdivisions are vulnerable too:** young developments with few mature trees can be the hottest places.
- **Action:** prioritize planting in the 27 shade deserts, require shade trees in new developments, and use LiDAR to monitor canopy over time.

Limitations: land surface temperature ≠ air temperature; 2018 LiDAR vs. 2018–22 heat & census; block-group averages (ecological fallacy); correlation, not causation.

References: Ziter et al. (2019) PNAS 116:7575 · Hoffman, Shandas & Pendleton (2020) Climate 8:12 · McDonald et al. (2021) PLOS ONE 16:e0249715 · Data: USGS 3DEP, USDA NAIP, USGS Landsat C2 L2, U.S. Census ACS, via Microsoft Planetary Computer.

Open code & data workflow
All Python scripts are reproducible from free data.
github.com/123sudhi/123sudhi.github.io
/tree/main/lidar-denton